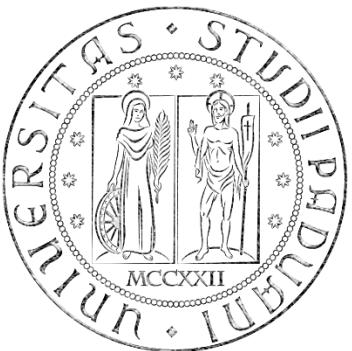


Biomedical Wearable Technologies for Healthcare and Wellbeing

Symposium

A.Y. 2024-2025

Giacomo Cappon



The devices



Polar H10 Heart Rate Monitor

- Bluetooth and ANT+ communication
- Sampling rate of 1 Hz

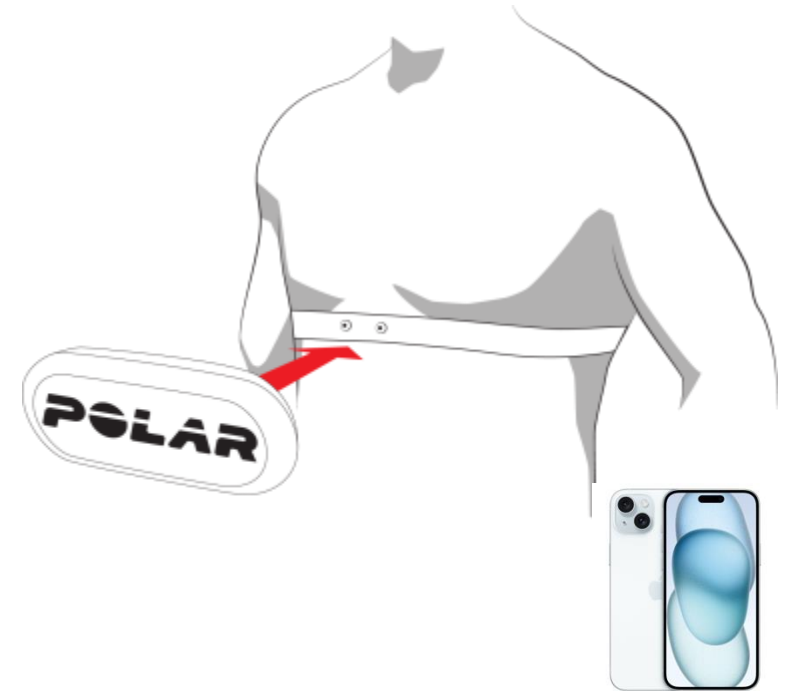
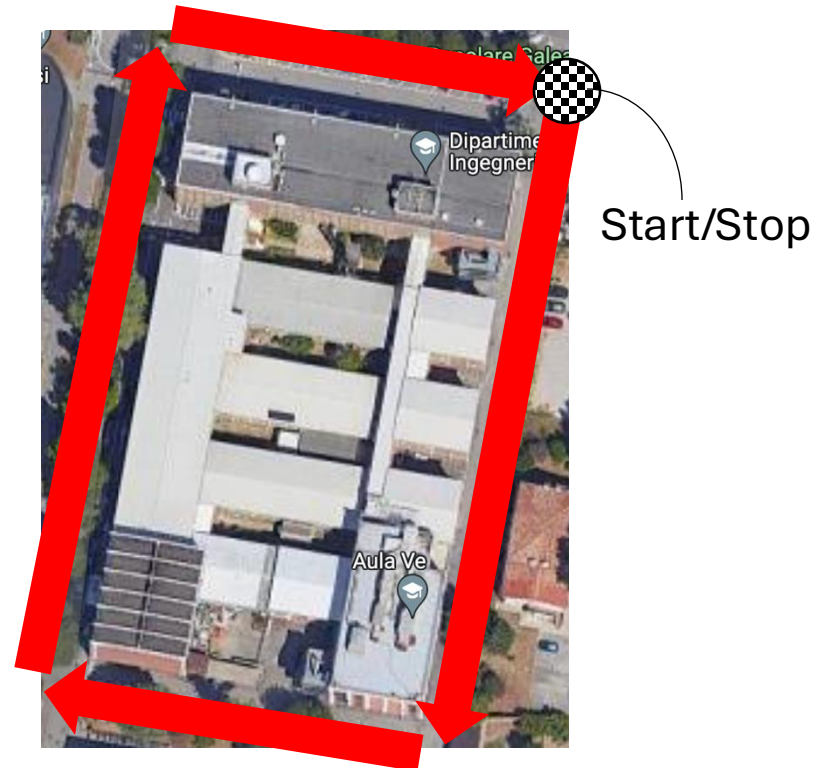


iPhone 15

- Hosts a custom app that communicates with Polar H10
- Collects and export data

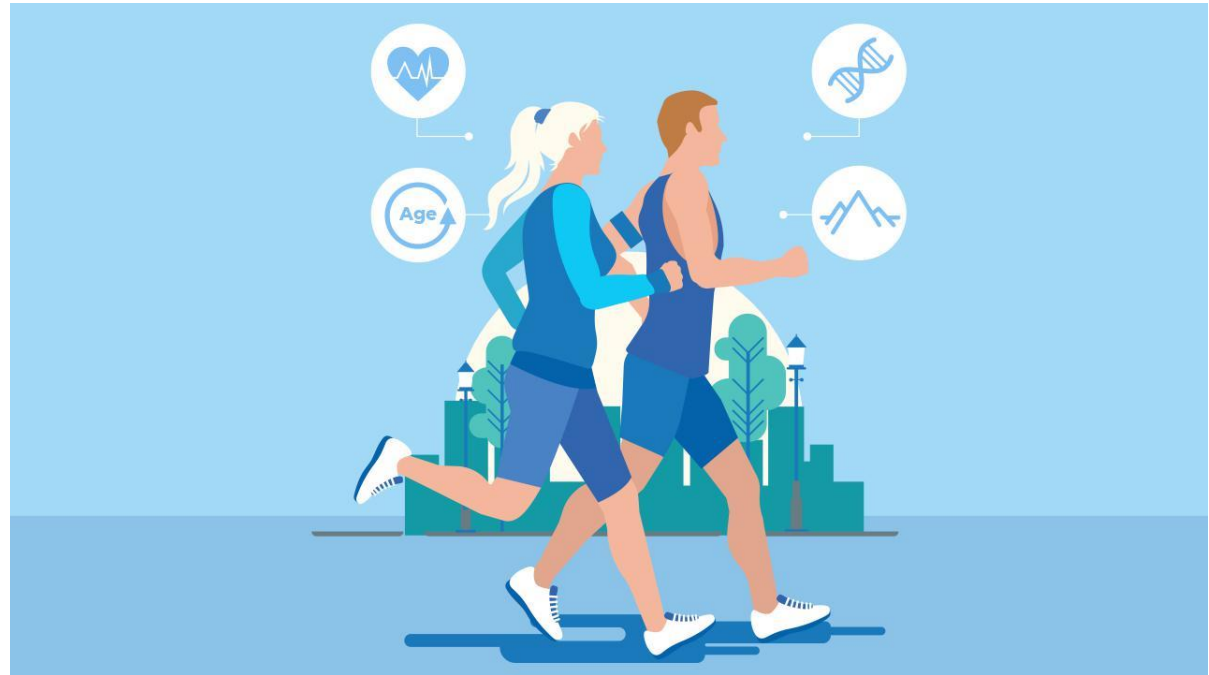
The “protocol”

- 3 volunteers wearing 1 x Polar H10 + 1x iPhone (need their Age and BW)
- Measure heart rate at rest
- 2 laps of the DEI “main” building running (or walking as fast as possible)
- At the end, export data



The aim

- Quantify and rank the activity “load” experienced by each participant



How data will look like

```
data =
```

```
147×2 table
```

hr	time
----	------

57	0
57	1
57	2
57	3
57	4
58	5
58	6
58	7
58	8
57	9

:	:
---	---

NaN	137
61	138
60	139
60	140
59	141
59	142
59	143
59	144
59	145
60	146

3 .csv files containing

- time from start (s)
- heart rate at given time (bpm)